Model Optimization and Tuning Phase Template

Date	14 april 2026
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Project Title	Restaurant Recommendation System
Maximum Marks	10 Marks

Model Optimization and Tuning Phase

The Model Optimization and Tuning Phase involves improving our machine learning recommendation model to get the best performance. This includes adjusting the model's parameters, experimenting with different algorithms, and selecting the most suitable model based on evaluation metrics such as accuracy, precision, recall, and RMSE (Root Mean Squared Error). Our restaurant recommendation system was designed to suggest similar restaurants based on location, user ratings, cuisines, and cost using collaborative filtering and content-based filtering techniques.

Hyperparameter Tuning Documentation (8 Marks):

Model	Tuned Hyperparameters
Model 1: Content-Based Filtering	- **Similarity Metric:** Cosine similarity was used as the primary metric to compute similarity between restaurants based on features like cuisines, rating, and cost. - **Top N Recommendations:** The number of top similar restaurants returned was tested with values like 5, 10, and 15.

```python
def recommend(name, cosine_similarities = cosine_similarities):

    # Create a list to put top restaurants
    recommend_restaurant = []

    # Find the index of the hotel entered
    idx = indices[indices == name].index[0]

    # Find the restaurants with a similar cosine-sim value and order them from bigges number
    score_series = pd.Series(cosine_similarities[idx]).sort_values(ascending=False)

    # Extract top 30 restaurant indexes with a similar cosine-sim value
    top30_indexes = list(score_series.iloc[0:31].index)

    # Names of the top 30 restaurants
    for each in top30_indexes:
        recommend_restaurant.append(list(df_percent.index)[each])

    # Creating the new data set to show similar restaurants
    df_new = pd.DataFrame(columns=['cuisines', 'Mean Rating', 'cost'])

    # Create the top 30 similar restaurants with some of their columns
    for each in recommend_restaurant:
        df_new = df_new.append(pd.DataFrame(df_percent[['cuisines','Mean Rating', 'cost']][df_percent.index == each].sample()))

    # Drop the same named restaurants and sort only the top 10 by the highest rating
    df_new = df_new.drop_duplicates(subset=['cuisines','Mean Rating', 'cost'], keep=False)
    df_new = df_new.sort_values(by='Mean Rating', ascending=False).head(10)

    print('TOP %s RESTAURANTS LIKE %s WITH SIMILAR REVIEWS: ' % (str(len(df_new)), name))
    df_new.index = df_new.index.str.lower()
    return df_new
```

Model 2: Collaborative Filtering

- **Algorithm:** SVD (Singular Value Decomposition) from the Surprise library.
- **Learning Rate:** Tuned values such as 0.005, 0.01, and 0.02 were tested.
- **Regularization:** Parameters such as 0.02, 0.05 were tried to avoid overfitting.
- **Number of Epochs:** Adjusted between 20 and 100 epochs.

```python
from surprise import SVD, Dataset, Reader
from surprise.model_selection import cross_validate

reader = Reader(rating_scale=(1, 5))
data = Dataset.load_from_df(df[['user_id', 'restaurant_name', 'rating']], reader)

svd = SVD()
cross_validate(svd, data, measures=['RMSE', 'MAE'], cv=5, verbose=True)
```

Final Model Selection Justification (2 Marks):

Final Model	Reasoning
Model 1: Content-Based Filtering	Selected due to its simplicity and good performance without requiring detailed user history. It gave interpretable and relevant results using restaurant features like cuisines, ratings, and cost.